

EDA: Univariate Analysis

Module 20 Study Notes • Exploratory Data Analysis

1. What is Univariate Analysis?

Exploratory Data Analysis (EDA) is the process of investigating data to discover patterns, anomalies, and test hypotheses using summary statistics and graphical representations. The most fundamental step in EDA is **Univariate Analysis**.

"Uni" means one, and "Variate" means variable. Therefore, Univariate Analysis is the process of examining one single column (variable) of your dataset at a time, completely independent of the others.

2. Types of Data

Before analyzing a column, you must determine its nature. Your visualization strategy depends entirely on whether the data is Categorical or Numerical.

- **Categorical Data:** Data that represents distinct groups or categories (e.g., Gender, Nationality, College Branch).
- **Numerical Data:** Data that represents measurable quantities (e.g., Age, Height, Salary).

3. Analyzing Categorical Data

For categorical data, the primary goal is to understand frequency and distribution.

- **Countplot (Bar Chart):** Displays the raw count (frequency) of each category. Use `sns.countplot(df['column_name'])` to quickly see which category dominates the dataset.
- **Pie Chart:** Displays the proportion or percentage of each category relative to the whole. Useful for understanding parts of a whole (e.g., what percentage of passengers were in 1st class vs 3rd class).

4. Analyzing Numerical Data

For numerical data, the goal is to understand the mathematical distribution, spread, and central tendency.

A. Histogram

A Histogram divides the numerical range into "bins" and counts how many data points fall into each bin. It is the best tool for visualizing the overall shape and spread of the data.

B. Distplot / KDE (Kernel Density Estimation)

Similar to a histogram, but it plots a smoothed probability density curve (PDF). The Y-axis represents the probability density. It is highly effective for identifying **Skewness**:

- **Symmetrical (Normal)**: The curve peaks perfectly in the middle.
- **Positive (Right) Skew**: The tail extends to the right (e.g., wealth distribution where a few people have massive salaries).
- **Negative (Left) Skew**: The tail extends to the left.

C. Boxplot

The Boxplot provides a visual representation of the **5-Number Summary**:

- **Minimum**: The lowest value excluding outliers.
- **Q1 (25th Percentile)**: 25% of the data falls below this line.
- **Median (Q2 / 50th Percentile)**: The exact middle value.
- **Q3 (75th Percentile)**: 75% of the data falls below this line.
- **Maximum**: The highest value excluding outliers.

Outlier Detection: Boxplots are the gold standard for spotting outliers. Any data point that falls outside the calculated Minimum or Maximum "whiskers" is plotted as an individual dot, indicating it is an anomaly that might need to be cleaned or removed.

5. Summary Statistics

Alongside visualizations, you should always check the exact numerical summaries using Pandas functions like `df['col'].min()`, `.max()`, `.mean()`, and `.skew()`.